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LASER MILLING SYSTEM FOR CONCENTRIC CASING STRING CEMENT REPAIR

Abstract

A method includes setting a primary plug within a wellbore flowpath defined through a radially innermost casing string of a plurality of concentric casing strings, operating a laser milling tool within the wellbore flowpath to discharge a laser beam into the wellbore flowpath to thereby mill a window through the radially innermost casing string and radially outward towards a radially innermost cement column of a plurality of respective cement columns surrounding each of the concentric casing strings, discharging an abrasive fluid from an abrasive jetting tool into the wellbore flowpath and cleaning out the radially innermost cement column to expand the window radially, flowing a secondary plug material into the window and the wellbore flowpath above the primary plug, solidifying the secondary plug material to form a solidified secondary plug within the window, and drilling out the solidified secondary plug to restore the wellbore flowpath through the solidified secondary plug.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates generally to concentric casing string cement repair and, more particularly, to methods and systems for laser milling of concentric casing strings for cement repair.

BACKGROUND OF THE DISCLOSURE

[0002] Oil and gas wellbores are commonly drilled in a series of progressively smaller casings until reaching a desired depth. A wellbore drilling operation may begin with drilling into a formation to a specified depth for a first casing string, also known as a first “casing depth”. The first casing string may be run downhole to the first casing depth and cemented in place by pumping cement between the formation and the first casing string to form a first stage cement column. The operation may continue with drilling to a second casing depth and running a second casing string downhole through the first casing string. The second casing string may then be cemented in place with a second stage cement column formed by pumping cement upward between the second casing string and the formation and continuing upward through a “casing-casing annulus” defined between the first casing string and the second casing string. The operation may continue with subsequent drilling and cementing stages until reaching a desired wellbore depth.

[0003] Once the drilling is complete, a production tubing may be installed within the innermost casing string, and production operations may be initiated to recover oil and gas resources through the production tubing. During the production operations, cracks or imperfections within the cement columns may lead to leaks or failures within the cement columns. These leaks may lead to a sustained casing pressure behind one or more casing strings, which may lead to undesirable flow within one or more casing-casing annuli and negatively affect overall wellbore integrity.

[0004] To avoid costly workover operations on wellbores with sustained casing pressure, methods have been developed to correct leaks or failures downhole. These methods include deploying a perforation gun or other tool to form perforations through the casing strings and cement columns, and then inserting a resin mixture within the perforated area to seal the leaks. Since the perforation gun may utilize explosives or hazardous equipment, forming the perforations may result in damage to the surrounding area and weakening of the geology surrounding the wellbore. Further, other repair methods may employ mechanical means for sectional milling or perforating of the casing strings and cement columns that employ full drilling rigs, and may therefore be costly and time-consuming.

[0005] Accordingly, methods and systems are desired for reliably correcting leaks and failures within concentric casings without mechanical milling means or a drilling rig.

SUMMARY OF THE DISCLOSURE

[0006] Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an exhaustive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

[0007] According to an embodiment consistent with the present disclosure, a method includes setting a primary plug within a wellbore flowpath defined through a radially innermost casing string of a plurality of concentric casing strings, operating a laser milling tool within the wellbore flowpath to discharge a laser beam into the wellbore flowpath to thereby mill a window through the radially innermost casing string and radially outward towards a radially innermost cement column of a plurality of respective cement columns surrounding each of the concentric casing strings. The method further includes discharging an abrasive fluid from an abrasive jetting tool into the wellbore flowpath and thereby cleaning out the radially innermost cement column to expand the window radially, flowing a secondary plug material into the window and the wellbore flowpath above the primary plug, solidifying the secondary plug material to form a solidified secondary plug within the window, and drilling out the solidified secondary plug to restore the wellbore flowpath through the solidified secondary plug.

[0008] In another embodiment, a wellbore repair system includes a plurality of concentric casing strings disposed within a wellbore, a plurality of cement columns disposed radially outward of each of the concentric casing strings, a window defined radially through at least a radially innermost casing string of the one or more of the concentric casing strings and a radially innermost cement column of the plurality of cement columns, wherein the window defines a first axial length through the radially innermost casing string and a second axial length through the radially innermost cement column, and wherein the second axial length is greater than the first axial length, a primary plug set within an interior of the radially innermost casing string below the window, and a solidified secondary plug filling the window above the primary plug.

[0009] In a further embodiment, a wellbore system includes a plurality of concentric casing strings disposed within a wellbore, a plurality of cement columns disposed radially outward of each of the concentric casing strings, a primary plug set within an interior of a radially innermost casing string of the plurality of concentric casing strings, a laser milled window defined in the radially innermost casing string and defining a first axial length, an abrasive jetting tool including a jetting head and one or more nozzles, couplable to the coiled tubing, and insertable within the flowpath of the wellbore to clean out cement from the radially innermost cement column to a second axial length greater than the first axial length by jetting an abrasive fluid behind the radially innermost casing string through the laser milled window, and a secondary plug material insertable into the window to generate a solidified secondary plug.

[0010] Any combinations of the various embodiments and implementations disclosed herein can be used in a further embodiment, consistent with the disclosure. These and other aspects and features can be appreciated from the following description of certain embodiments presented herein in accordance with the disclosure and the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a schematic cross-sectional side view of a wellbore with sustained casing pressure within concentric casing strings, which may be repaired with systems and methods according to one or more embodiments of the present disclosure.

[0012] FIG. **2** is a schematic cross-sectional side view of the wellbore with production tubing removed, a primary plug installed and a laser milling tool inserted therein for progressive milling of the concentric casing strings, according to an initial step of a repair operation of one or more embodiments of the present disclosure.

[0013] FIG. **3** is a schematic cross-sectional side view of the laser milling tool, according to one or more embodiments of the present disclosure.

[0014] FIG. **4** is a schematic cross-sectional side view of the wellbore with an abrasive fluid injection tool inserted therein for progressive removal of cement, according to subsequent steps of the repair operation.

[0015] FIG. **5** is a schematic cross-sectional side view of the wellbore with a completed window partially filled with a secondary plug material, according to subsequent steps of the repair operation.

[0016] FIG. **6** is a schematic cross-sectional side view of the wellbore with a solidified secondary plug, according to subsequent steps of the repair operation.

[0017] FIG. **7** is a schematic cross-sectional side view of the wellbore with a drill string inserted therein for drilling through the primary and secondary plug, according to subsequent steps of the repair operation.

[0018] FIG. **8** is a schematic cross-sectional side view of the wellbore with production tubing reinstalled, according to subsequent steps of the repair operation.

[0019] FIG. **9** is a schematic flowchart of an example method for correcting a leak within concentric casing strings via a laser milling system.

DETAILED DESCRIPTION

[0020] Embodiments of the present disclosure will now be described in detail with reference to the accompanying Figures. Like elements in the various figures may be denoted by like reference numerals for consistency. Further, in the following detailed description of embodiments of the present disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the claimed subject matter. However, it will be apparent to one of ordinary skill in the art that the embodiments disclosed herein may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description. Additionally, it will be apparent to one of ordinary skill in the art that the scale of the elements presented in the accompanying Figures may vary without departing from the scope of the present disclosure.

[0021] Embodiments in accordance with the present disclosure generally relate to concentric casing string cement repair and, more particularly, to methods and systems for concentric casing string cement repair using laser milling tools that do not require a full drilling rig. The embodiments disclosed herein include methods and systems which utilize a laser-milled and abrasively jetted window extending through concentric casing strings and cement columns to approach a leak. The methods and systems may further involve introducing a primary plug for milling and filling operations, which may include introducing a secondary plug into the wellbore flowpath above the primary plug to fill in any leaks or failures. The secondary plug may form a gas-tight seal within and around the leaks or failures. The primary and secondary plugs may be further milled out to restore a wellbore flowpath downhole, such that further operations may continue in the wellbore once the leaks or failures are corrected. Accordingly, the methods and systems disclosed herein may enable rapid equipment deployment for the sealing of leaks causing sustained casing pressure without requiring the use of a full drilling rig. Progressive laser milling and abrasively jetted cement cleaning may enable the sealing of leaks within the outermost concentric casing strings or cement columns without requiring full workover operations.

[0022] FIG. **1** is a schematic cross-sectional side view of a wellbore **100** with sustained casing pressure behind one or more of a plurality of concentric casing strings **102***a-c* disposed within the wellbore **100**, according to one or more embodiments of the present disclosure. A fist casing-casing annulus CCA-**1** is defined radially between casing strings **102***c* and **102***b*, and a second casing-casing annulus CCA-**2** is defined between casing strings **102***a*, **102***b*. The sustained casing pressure in the illustrated embodiment may be due to a failure in a cement column **104***a* around the first casing string **102***a*. The cement column **104***a*, as shown, includes a plurality of leaks **106** which may enable flow within the cement column **104***a*. In the illustrated embodiment, the leaks **106** further penetrate through the first casing string **102***a* due to the sustained casing pressure and associated damage. In some embodiments, however, the sustained casing pressure may be caused by channels within the cement column **104***a* in fluid communication with a surface location (e.g.,) without damage to the first casing string **102***a*. To repair the leaks **106** behind (radially outward of) the first casing string **102***a*, corrective operations disclosed herein may include selective removal of a portion of second and third casing strings **102***b*,*c* and cement columns **104***b*,*c* between the wellbore flowpath **108** and the leaks **106**. However, any of the cement columns **104***a-c* and any of the first, second, or third casing strings **102***a-c* may include leaks or failures that may cause sustained casing pressure within the wellbore **100**. Prior to, or in concert with, a beginning of repair operations within the wellbore **100**, any production tubing **110** inserted therein may be retracted out of the wellbore **100**.

[0023] Example progressive steps of a repair operation will now be provided with reference to FIGS. **2** and **4-8**, which depict a series of cross-sectional side views of the wellbore **100**, according to one or more embodiments.

[0024] FIG. **2** is a schematic cross-sectional side view of the wellbore **100** with production tubing **110** (FIG. **1**) removed from the wellbore flowpath **108**, and a laser milling tool **200** inserted therein according to an initial step of the repair operation of one or more embodiments of the present disclosure. The laser milling tool **200** may be inserted into the wellbore flowpath **108** following setting of a primary plug **202** within the third casing string **102***c*, which may be referred to as the radially innermost casing string. The primary plug **202** may be set within the wellbore flowpath **108** and may isolate any lower portions of the wellbore flowpath **108** from the area surrounding the leaks **106**. In some embodiments, the primary plug **202** may be set a certain distance below the leaks **106** as illustrated to enable correction in the local area of the leaks **106**. In other embodiments, the primary plug **202** may be set above the leaks **106** to seal leak paths extending between the leaks **106** and a surface location. In some embodiments, the primary plug **202** may be a bridge plug and may be either retrievable or drillable/millable in nature.

[0025] The laser milling tool **200** may be run downhole on coiled tubing **206** or another wellbore conveyance. The coiled tubing **206** may enable transfer of a laser beam **309** (FIG. **3**) or laser energy from the surface to the laser milling tool **200** downhole. An operator at the surface may utilize a coiled tubing unit (not shown) on the surface to raise and lower the coiled tubing **206** for deploying, positioning, maneuvering, and withdrawing the laser milling tool **200** without a full drilling rig, as will be appreciated by those skilled in the art. The laser milling tool **200** may include a laser head **210** and a support structure **208** for coupling the laser head **210** to the coiled tubing **206**.

[0026] The laser head **210** may include a body **212** with a nozzle **214** protruding therefrom. The body **212** may protect internal components of the laser head **210** (see FIG. **3**) from the ambient environment downhole, while the nozzle **214** may provide a path for laser beam **309** (FIG. **3**) to reach the third (innermost) casing string **102***c*. The laser milling tool **200** may be translated and rotated from the coiled tubing unit on the surface, or may be locally translated via motors (not shown) within the laser milling tool **200**. Translation and rotation of the laser milling tool **200** and nozzle **214** may enable selective milling of the third casing string **102***c* in the illustrated pattern. Accordingly, the milling of the third casing string **102***c* in the illustrated pattern may begin formation of a window **204**, in which a secondary plug **600** (FIG. **6**) may later be installed as described below.

[0027] In some embodiments, the laser milling tool **200** may be operated in an environment filled with an optical fluid “OF” selected to facilitate transmitting the laser energy within the wellbore **100**. In these embodiments, the optical fluid “OF” may be stored at a first external location **216** (e.g., at the surface or otherwise outside the wellbore **100**). The optical fluid “OF” may be of a density to maintain well control within the wellbore **100** while possessing optical properties to enable laser emission therethrough. In some embodiments, the optical fluid “OF” may include liquids such as glycerin, glycols, or alcohols. In other embodiments the optical fluid “OF” may include gaseous fluids such as nitrogen or argon, or other inert gases. The optical fluid “OF” may be provided to the wellbore **100** via a fluid line **218** which is in further communication with a pump **220**. The pump **220** may provide the optical fluid “OF” at sufficient volume and flowrate to replace any downhole fluids with the optical fluid “OF” within the area surrounding the window **204**. In some embodiments, the pump **220** is in fluid communication with a reservoir **222**, which may be a fluid tank, repurposed wellbore, or other fluid container for storing optical fluid “OF”.

[0028] The laser milling tool **200** may be operated within the wellbore **100** until the desired window **204** has been milled from the third casing string **102***c*. In some embodiments, the laser milling tool **200** may be pulled out of hole to enable deployment of further tooling for cleaning out the cement column **104***c*. In further embodiments, however, the laser milling tool **200** may be utilized in both milling of the third casing string **102***c* and the third cement column **104***c*, without departing from the scope of this disclosure.

[0029] FIG. **3** is a schematic cross-sectional side view of the laser milling tool **200**, according to one or more embodiments of the present disclosure. As discussed above, the laser milling tool **200** be run downhole on, coiled tubing **206**. The coiled tubing **206** may include an insulated fiber optic cable **301** therein for transmitting the laser beam **309** and/or laser energy downhole. The laser milling tool **200** may be fixedly coupled to the coiled tubing **206** via the support structure **208** as described above. The support structure **208** may further include a support rod **302** and a plurality of support rings **304**. The support rod **302** may be fixedly coupled to the body **212** and/or nozzle **214** of the laser head **210** and may provide a structural backbone for the laser milling tool **200**. The support rod **302** supports the plurality of support rings **304** at axially spaced locations along the fiber optic cable **301**. The support rings **304** may be sized to receive, clamp or otherwise secure the fiber optic cable **301**, such that the support rod **302** fixes an end of the fiber optic cable **301** at a desired position and orientation with respect to the laser head **210**. The support structure **208** may thus transmit rotational and longitudinal motion from the coiled tubing **206** to the laser head **210** without disrupting transmission of the laser beam **309** to the laser head **210**.

[0030] The nozzle **214** of the laser milling tool **200** may extend into the body **212** to receive the laser beam **309** or energy from the coiled tubing **206** and/or fiber optic cable **301**. The nozzle **214** may include a reflector **306** therein for angular redirection of the laser beam **309** towards an end of the nozzle **214**. The reflector **306** may include an angled mirror, beam splitter, or prism capable of aiming the laser beam **309** towards a desired target while controlling the orientation, size, and number of beams produced within the nozzle **214**. Further, the nozzle **214** may include a focus lens **308** interposing an outlet end of the nozzle **214** and the reflector **306**, such that the redirected laser beam **309** from the reflector **306** may be appropriately focused for milling operations on a target exterior to the nozzle **214**. In the illustrated embodiment, the focus lens **308** alters the shape of the laser beam **309** to focus the laser beam **309** to a focal point **311** within the nozzle **214**, such that an output defocused beam **310** may divergently exit the nozzle **214** towards a target. In other embodiments, the defocused beam **310** may be collimated with a collimator (not shown) before exiting the nozzle **214**.

[0031] The focus lens **308** may be supported within the nozzle **214** via one or more lens supports **312**. The one or more lens supports **312** may couple the focus lens **308** to the nozzle **214** while retaining the focus lens **308** in a particular location or orientation. In the illustrated embodiment, the lens supports **312** include a fluid knife generator **314** therein. The fluid knife generator **314** may expel air, specific gases, or optical fluids into the nozzle **214** for protecting the focused lens **308** and other internal components from debris and external fluids. The fluid knife generator **314** may expel a pressurized optical fluid within or near the nozzle **214** to isolate the nozzle **214** from the wellbore flowpath **108** and any generated debris. In some embodiments, a thin curtain of fluid may be continuously provided within the nozzle **214** to prevent further fluids or debris from entering the nozzle **214**. In further embodiments, however, the fluid knife generator **314** may pump expel pressurized fluid into the nozzle **214** at a greater pressure than the wellbore environment to prevent flow into the nozzle **214**. The fluid knife generator **314** may be located at or near the outlet end of the nozzle and may provide an unobstructed area within the nozzle **214** through which the defocused beam **310** may pass.

[0032] FIG. **4** is a schematic cross-sectional side view of the wellbore **100** with an abrasive fluid injection tool **400** inserted therein for progressive removal of cement from portions of the cement columns **104***a*, **104***b*, etc. according to subsequent steps of the repair operation. Following milling of the window **204** within the third (innermost) casing string **102***c*, the abrasive fluid injection tool

400 may be positioned within the wellbore flowpath **108** to expand the window **204** radially outward through the third cement column **104***c*. Operation of the abrasive fluid injection tool **400** may include running an injection head **402** downhole on coiled tubing **206** until the injection head **402** reaches the window **204**. An abrasive fluid “AF” may be provided through the coiled tubing **206** to and injected into the window **204** via one or more injection nozzles **404** of the injection head **402**. The abrasive fluid “AF” is directed towards the cement columns **104***b-c* to clean out exposed cement. The abrasive fluid “AF” may be provided to the coiled tubing **206** from a second external location **406** (external to the wellbore **100** via a fluid line **408**. The fluid line **408** may be in fluid communication with a pump **410**, which may provide the abrasive fluid “AF” at sufficient volume and flowrate to effectively clean out the exposed cement. In some embodiments, the pump **410** is in fluid communication with a reservoir **412**, which may be a fluid tank, repurposed wellbore, or other fluid container for storing abrasive fluid “AF”. In some embodiments, the abrasive fluid “AF” may be a suspension of sand within a working fluid, such as water with a thickening agent, which may be used in cement cleanout tasks. The abrasive fluid injection tool **400** may accordingly clean out the cement columns **104***b-c* until reaching an inner surface of a surrounding casing string, e.g., the second casing string **102***b* as illustrated in FIG. **4**. The expansion of the window **204** via the abrasive fluid injection tool **400** may expose an inner surface of the second casing string **102***b* to enable further laser milling operations to be conducted on the second casing string **102***b* once the abrasive fluid injection tool **400** is withdrawn from the wellbore. In further embodiments, the abrasive fluid injection tool **400** may include one or more scrapers (not shown) with blades or brushes formed of steel to mechanically remove the cement columns **104***b-c*.

[0033] Further operations of the laser milling tool **200** and abrasive fluid injection tool **400** may be performed as needed to further radially expand the window **204** and thereby reach the location of the leaks **106**. In some embodiments, new optical fluid “OF”, or a further cleaning fluid, may be pumped downhole after milling and cement cleanout (abrasive jetting) operations to remove any debris and dust from the previous operations. In the illustrated embodiment, the leaks **106** are located within the first cement column **104***a* and the first casing string **102***a*, and the illustrated operations may be performed until exposing the first casing string **102***a* through the second cement column **104***b*. While three concentric casing strings **102***a-c* are illustrated here, the laser milling tool **200** and abrasive fluid injection tool **400** may be deployed through any number of casing strings **102***a-c* without departing from the scope of this disclosure. In some embodiments, the third casing string **102***c* may be a radially innermost casing string, while the second casing string **102***b* may be a radially outward casing string circumscribing the radially innermost casing string, agnostic of the number of casing strings **102***a-c* present within the wellbore **100**. In the illustrated embodiment, the laser milling and abrasive jetting operations may been performed up until the window **204** is in fluid communication with the leaks **106** (see FIG. **5**).

[0034] FIG. **5** is a schematic cross-sectional side view of the wellbore **100** with a completed window **204** partially filled with a secondary plug material **500**, according to subsequent steps of the repair operation. Following the laser milling and abrasive jetting operations discussed above, coiled tubing **206** may be inserted into the wellbore flowpath **108** above the window **204**. The coiled tubing **206** may deliver the secondary plug material **500** into the window **204** to create a secondary plug **600** (FIG. **6**) above the primary plug **202**. The secondary plug material **500** may include a resin mixture, a cement mixture, a molten eutectic alloy, a solid eutectic alloy to be melted downhole, or a combination thereof. The secondary plug material **500** may fill the window **204** and may seep into the leaks **106** present therein, such that the leaks **106** may be plugged.

[0035] In the illustrated embodiment, the window **204** defines a plurality of axial lengths L.sub.1-L.sub.4 are shown within the window **204**. The plurality of lengths L.sub.1-L.sub.4 may be generated via the laser milling operations and abrasive jetting operations of FIGS. **2** and **4**. As shown in the illustrated embodiment, the length L.sub.2 of the cleaned out cement from the third cement column **104***c* may be greater than the length Li removed from the third casing string **102***c*.

The laser milling operations of FIG. **2** may be utilized in generating the length L.sub.1 within the third casing string **102***c*, while the abrasive jetting operations of FIG. **4** may be utilized in generating the length L.sub.2 within the third cement column **104***c* up and behind the third casing string **102***c*. Further, the length L.sub.3 removed from the second casing string **102***b* may be less than length L.sub.1 removed from the third casing string **102***c*. Similarly, the length L.sub.4 removed from the second cement column **104***b* may be less than length L.sub.2 removed from the third cement column **102***b*. The stepped nature of the removed casing strings **102***a-c* and cement columns **104***a-c* may enable enhanced sealing with the secondary plug material **500** within the window **204**. The exposed casing strings **102***a-c* may be bonded to the secondary plug material **500**, such that the secondary plug material **500** may be held in place even following drilling or milling of the secondary plug material **500** (see FIG. **7**).

[0036] FIG. **6** is a schematic cross-sectional side view of the wellbore **100** with a solidified secondary plug **600**, according to subsequent steps of the repair operation. As shown in the illustrated embodiment, the secondary plug material **500** (FIG. **5**) has hardened into a solidified secondary plug **600** which has expanded to fill the window **204**, any micro-annuli forming leak paths, the leaks **106**, and the wellbore flowpath **108** above the primary plug **202**. Depending upon the type of secondary plug material **500** introduced into the window **204**, a squeezing tool **602** may be utilized in setting the solidified secondary plug **600**. The squeezing tool **602** may be pressed down upon the secondary plug material **500** during solidification to aid in maintaining a shape and penetration of the secondary plug material **500** into the leaks **106**.

[0037] The solidified secondary plug **600** may form a gas-tight seal within any voids present in the wellbore **100** above the primary plug **202**. In some embodiments, the solidified secondary plug **600** may bond to the casing strings **102***a-c* at exposed surfaces within the axial lengths L.sub.1-L.sub.4 (FIG. **5**) of the window **204**. In these embodiments, a seal may be created within the wellbore **100**, such that a quality gas-tight seal is present between the casing strings **102***a-c* and the solidified secondary plug **600**. Accordingly, any leaks **106** or other failures may be filled or plugged, such that any sustained casing pressure issues may be remediated. In some embodiments, the secondary plug material **500** includes a eutectic alloy that may expand during solidification, and may provide an additional benefit of generating a metal-to-metal seal with the casing strings **102***a-c* for further sealing and reliability.

[0038] FIG. **7** is a schematic cross-sectional side view of the wellbore **100** with a drill string **700** inserted within the wellbore flowpath **108**, according to one or more embodiments of the present disclosure. The drill string **700** may include a drill bit **702** therein that is chosen to drill through the solidified secondary plug **600**, whether that is cement, resin, or metallic alloy. In some embodiments, the drill bit **702** may be a milling element for milling through the solidified secondary plug **600**. The drill string **700** may be advanced within the wellbore flowpath **108** until the drill bit **702** reaches the solidified secondary plug **600**. The drill string **700** may be in communication with an external hydraulic motor **704** configured to provide a desired motion to the drill bit **702** without the use of a full drilling rig.

[0039] As shown in the illustrated embodiment, the drill bit **702** may then be utilized in drilling out the wellbore flowpath **108** through both the solidified secondary plug **600** and the primary plug **202**. In some embodiments, the primary plug **202** may be a retrievable bridge plug set within the wellbore flowpath **108**. In these embodiments, the drill bit **702** may drill out the solidified secondary plug **600** up to the primary plug **202**, at which point the primary plug **202** may be unset and retracted out of the wellbore **100**. Regardless of the type of primary plug **202** utilized, the wellbore flowpath **108** may be restored through the repaired area to enable further use of the wellbore **100**. The drill bit **702** may be chosen to match the diameter of the third casing string **102***c*, such that the wellbore flowpath **108** may remain constantly sized throughout the wellbore **100**.

[0040] FIG. **8** is a schematic cross-sectional side view of the wellbore **100** with the production tubing **110** reinstalled, according to one or more embodiments of the present disclosure. The production tubing **110** may be reinstalled within the wellbore flowpath **108** through the solidified secondary plug **600** after drilling. The remaining portions of the solidified secondary plug **600** may maintain a seal within the leaks **106**, such that the sustained casing pressure may be alleviated or removed. The production tubing **110** may then be reutilized in hydrocarbon production operations within the wellbore **100** without sustained casing pressure.

[0041] Through the progressive utilization of the laser milling system as shown in FIGS. **2** and **4-8**, the sustained casing pressure may be remediated within the concentric casing strings **102***a-c* and cement columns **104***a-c*. The laser milling system may include, but is not limited to, the primary plug **202**, the laser milling tool **200**, the abrasive fluid injection tool **400**, the solidified secondary plug **600**, the drill string **700**, and any components thereof that are utilized in the embodiments illustrated herein.

[0042] FIG. **9** is a schematic flowchart of an example method **900** for correcting a leak (e.g., the leaks **106**) within concentric casing strings (e.g., concentric casing strings **102***a-c*) via a laser milling system. The method **900** may include setting a primary plug (e.g., the primary plug **202**) within the wellbore flowpath (e.g., the wellbore flowpath **108**) near, above or below the location of the leaks at **902**. The setting of the primary plug at **902** may enable further repair processes to be performed within the wellbore (e.g., the wellbore **100**) near the leaks without affecting the remainder of the wellbore flowpath. The primary plug may be a drillable plug which may be drilled out at a later time, or may be an expandable, retrievable plug which may be collapsed and retracted out of hole following repairs.

[0043] The method **900** may include laser milling out a portion of a concentric casing string via a laser milling tool (e.g., the laser milling tool **200**) at **904**. The laser milling of the concentric casing string may create a window (e.g., the window **204**) within the concentric casing string. In some embodiments, the laser milling tool may be run downhole on coiled tubing (e.g., the coiled tubing **206**) such the method **900** may be completed without a drilling rig. The window milled out at **904** may enable access to one or more cement columns (e.g., the one or more cement columns **104***a-c*) within the wellbore flowpath.

[0044] The method **900** may further include cleaning out one or more cement columns behind the concentric casing string at **906** via an abrasive jetting tool (e.g., the abrasive fluid injection tool **400**). The abrasive jetting tool may be deployable within the wellbore flowpath to expand the window through one or more cement columns to provide access to the leaks or another concentric casing string for further operations. In some embodiments, the abrasive jetting tool may be run downhole on coiled tubing such that a drilling rig may not be required for completing the method **900**. In some embodiments, three or more concentric casing strings may be installed within the wellbore. In these embodiments, based on the locations of the leaks, the laser milling at **904** and abrasive jetting cement cleaning at **906** may be repeated in progressive operations until the leaks are in fluid communication with the window and wellbore flowpath. In some embodiments, the laser milling at **904** may continue into the cement columns, such that the cement is milled out via the laser milling tool. In these embodiments, the abrasive jetting tool may remove any remaining cement in the milled area. In some embodiments, the abrasive jetting tool may further include one or more steel blades or brushes to aid in cleaning out the cement columns.

[0045] The method may further include flushing out the window with a clean fluid (e.g., the optical fluid "OF") to remove any further debris or dust at **908**. The clean fluid may circulate within the window and may carry out any contaminants, such as the debris or dust, to enable further operations in a clean environment. The flushing of the window at **908** may be performed any number of times between downhole operations as subsequent millings or cement cleanouts are performed. In some embodiments, cleaning out the cement columns may expose interior surfaces of one or more concentric casing strings. In these embodiments, the abrasive jetting operations at **906** may expand the window of cleaned cement behind the concentric casing string to a length greater than that of the window through the concentric casing string.

[0046] The method **900** may further include inserting a secondary plug material (e.g., the secondary plug material **500**) into the milled and cleaned out window at **910**. The secondary plug material may include a resinous material, a cement material, or a eutectic alloy for formation of a secondary plug within the wellbore flowpath. The secondary plug material may form a bond with the exposed cement columns and/or concentric casing strings to generate a seal within the window and wellbore flowpath. The method **900** may continue at **912** with solidifying the secondary plug material into a solidified secondary plug (e.g., the solidified secondary plug **600**). The solidification of the secondary plug may be facilitated through a squeezing tool (e.g., the squeezing tool **602**) for resins or cements, or may be completed through melting and cooling of a eutectic alloy. The solidified secondary plug may penetrate the leaks within the wellbore and may generate a gas-tight seal within the wellbore flowpath to repair the cement columns and casing strings.

[0047] The method **900** may further include drilling out the wellbore flowpath through the solidified secondary plug at **914** via a drill string (e.g., the drill string **700**). The drill string may include a drill bit (e.g., the drill bit **702**) installed thereon for drilling out of the solidified eutectic plug in the same diameter as the wellbore flowpath. In some embodiments, the drill string **700** may be powered by a hydraulic motor (e.g., the external hydraulic motor **704**) such that a drilling rig may not be necessary to complete the method **900**. In some embodiments, the drilling out of the wellbore flowpath at **914** may include drilling through the plug previously set at **902**. In further embodiments, however, the method **900** can include unsetting the plug and retracting the plug out of hole at **916** for embodiments utilizing expandable, retrievable plugs.

[0048] The method **900** may further include running production tubing (e.g., the production tubing **110**) within the wellbore flowpath including the leak remediation therein at **916**. The running of production tubing **110** may enable further wellbore operations within the wellbore without sustained casing pressure, such that normal operations of the wellbore may continue.

[0049] Embodiments disclosed herein include:

[0050] A. A method comprising setting a primary plug within a wellbore flowpath defined through a radially innermost casing string of a plurality of concentric casing strings, operating a laser milling tool within the wellbore flowpath to discharge a laser beam into the wellbore flowpath to thereby mill a window through the radially innermost casing string and radially outward towards a radially innermost cement column of a plurality of respective cement columns surrounding each of the concentric casing strings, discharging an abrasive fluid from an abrasive jetting tool into the wellbore flowpath and thereby cleaning out the radially innermost cement column to expand the window radially, flowing a secondary plug material into the window and the wellbore flowpath above the primary plug, solidifying the secondary plug material to form a solidified secondary plug within the window, and drilling out the solidified secondary plug to restore the wellbore flowpath through the solidified secondary plug.

[0051] B. A wellbore repair system comprising a plurality of concentric casing strings disposed within a wellbore, a plurality of cement columns disposed radially outward of each of the concentric casing strings, a window defined radially through at least a radially innermost casing string of the one or more of the concentric casing strings and a radially innermost cement column of the plurality of cement columns, wherein the window defines a first axial length through the radially innermost casing string and a second axial length through the radially innermost cement column, and wherein the second axial length is greater than the first axial length, a primary plug set within an interior of the radially innermost casing string below the window, and a solidified secondary plug filling the window above the primary plug.

[0052] C. A wellbore system comprising a plurality of concentric casing strings disposed within a wellbore, a plurality of cement columns disposed radially outward of each of the concentric casing strings, a primary plug set within an interior of a radially innermost casing string of the plurality of concentric casing strings, a laser milled window defined in the radially innermost casing string and defining a first axial length, an abrasive jetting tool including a jetting head and one or more nozzles, couplable to the coiled tubing, and insertable within the flowpath of the wellbore to clean out cement from the radially innermost cement column to a second axial length greater than the first axial length by jetting an abrasive fluid behind the radially innermost casing string through the laser milled window, and a secondary plug material insertable into the window to generate a solidified secondary plug

[0053] Each of embodiments A through C may have one or more of the following additional elements in any combination: Element 1: further comprising: drilling into the primary plug subsequent to drilling out the solidified secondary plug to restore the wellbore flowpath, wherein the primary plug is a drillable plug. Element 2: further comprising: releasing the primary plug from below the drilled solidified secondary plug; and retracting the primary plug from the wellbore flowpath, wherein the primary plug is an expandable, releasable plug. Element 3: further comprising: running a production tubing through the solidified secondary plug subsequent to restoring the wellbore flowpath therethrough. Element 4: further comprising: pumping an optical fluid into the wellbore flowpath; and transmitting the laser beam through the optical fluid. Element 5: wherein the window extends through the radially innermost string at a first axial length, and wherein cleaning out the radially innermost cement column axially extends the window through the radially innermost cement column to a second axial length that is greater than the first axial length. Element 6: further comprising: milling through a radially outward casing string of the plurality of casing strings circumscribing the radially innermost casing string to expand the window radially outwardly through the radial outward casing string. Element 7: wherein milling through the radially outward casing string axially extends the window to a third length shorter than that of the first length and the second length. Element 8: further comprising: cleaning out the respective cement column surrounding the radially innermost casing string to further expand the window radially outwardly. Element 9: wherein operating the laser milling tool further includes: expelling a pressurized optical fluid within or near an outlet of the laser beam and thereby isolating the outlet of the laser beam from the wellbore flowpath and debris.

[0054] Element 10: wherein the window is further defined radially through a casing string radially outward of the radially innermost casing string to a third axial length and through a cement column radially outward of the radially innermost cement column to a fourth axial length. Element 11: wherein the second axial length is greater than that of the third axial length and the fourth axial length, and wherein the first axial length is greater than that of the third axial length. Element 12: further comprising a window defined radially through at least the radially innermost casing string to a first axial length. Element 13: wherein the window is further defined through the radially innermost cement column to a second axial length, and wherein the second axial length is greater than the first axial length. Element 14: wherein the laser milling tool further includes: a reflector within the nozzle and oriented to redirect a laser beam from the coiled tubing towards an outlet end of the nozzle; and a focus lens interposing the outlet end of the nozzle and the reflector, and operable to focus the laser beam through the outlet end of the nozzle. Element 15: further comprising: a fluid reservoir stored at an external location; a pump in fluid communication with the fluid reservoir; and a fluid line in fluid communication with the fluid reservoir and the coiled tubing or the flowpath of the wellbore, wherein the fluid reservoir stores fluid for use in laser milling or abrasive jetting operations. Element 16: wherein the fluid is an optical fluid for flushing of the wellbore and transmission of a laser beam therethrough. Element 17: wherein the fluid is an abrasive fluid for cleaning out cement via the abrasive jetting tool.

[0055] By way of non-limiting example, exemplary combinations applicable to A through C include: Element 5 with Element 6; Element 5 with Element 7; Element 5 with Element 8; Element 10 with Element 11; Element 12 with Element 13; Element 15 with Element 16; Element 15 with Element 17.

[0056] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, for example, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “contains”, “containing”, “includes”, “including,” “comprises”, and/or “comprising,” and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0057] Terms of orientation used herein are merely for purposes of convention and referencing and are not to be construed as limiting. However, it is recognized these terms could be used with reference to an operator or user. Accordingly, no limitations are implied or to be inferred. In addition, the use of ordinal numbers (e.g., first, second, third, etc.) is for distinction and not counting. For example, the use of “third” does not imply there must be a corresponding “first” or “second.” Also, if used herein, the terms “coupled” or “coupled to” or “connected” or “connected to” or “attached” or “attached to” may indicate establishing either a direct or indirect connection, and is not limited to either unless expressly referenced as such.

[0058] While the disclosure has described several exemplary embodiments, it will be understood by those skilled in the art that various changes can be made, and equivalents can be substituted for elements thereof, without departing from the spirit and scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation, or material to embodiments of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiments disclosed, or to the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims. Moreover, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, or component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative.

Claims

1. A method comprising: setting a primary plug within a wellbore flowpath defined through a radially innermost casing string of a plurality of concentric casing strings; operating a laser milling tool within the wellbore flowpath to discharge a laser beam into the wellbore flowpath to thereby mill a window through the radially innermost casing string and radially outward towards a radially innermost cement column of a plurality of respective cement columns surrounding each of the concentric casing strings; discharging an abrasive fluid from an abrasive jetting tool into the wellbore flowpath and thereby cleaning out the radially innermost cement column to expand the window radially; flowing a secondary plug material into the window and the wellbore flowpath above the primary plug; solidifying the secondary plug material to form a solidified secondary plug within the window; and drilling out the solidified secondary plug to restore the wellbore flowpath through the solidified secondary plug.

2. The method of claim 1, further comprising: drilling into the primary plug subsequent to drilling out the solidified secondary plug to restore the wellbore flowpath, wherein the primary plug is a drillable plug.

3. The method of claim 1, further comprising: releasing the primary plug from below the drilled solidified secondary plug; and retracting the primary plug from the wellbore flowpath, wherein the primary plug is an expandable, releasable plug.

4. The method of claim 1, further comprising: running a production tubing through the solidified secondary plug subsequent to restoring the wellbore flowpath therethrough.

5. The method of claim 1, further comprising: pumping an optical fluid into the wellbore flowpath;

and transmitting the laser beam through the optical fluid.

6. The method of claim 1, wherein the window extends through the radially innermost string at a first axial length, and wherein cleaning out the radially innermost cement column axially extends the window through the radially innermost cement column to a second axial length that is greater than the first axial length.

7. The method of claim 6, further comprising: milling through a radially outward casing string of the plurality of casing strings circumscribing the radially innermost casing string to expand the window radially outwardly through the radial outward casing string.

8. The method of claim 7, wherein milling through the radially outward casing string axially extends the window to a third length shorter than that of the first length and the second length.

9. The method of claim 7, further comprising: cleaning out the respective cement column surrounding the radially innermost casing string to further expand the window radially outwardly.

10. The method of claim 1, wherein operating the laser milling tool further includes: expelling a pressurized optical fluid within or near an outlet of the laser beam and thereby isolating the outlet of the laser beam from the wellbore flowpath and debris.

11. A wellbore repair system, comprising: a plurality of concentric casing strings disposed within a wellbore; a plurality of cement columns disposed radially outward of each of the concentric casing strings; a window defined radially through at least a radially innermost casing string of the one or more of the concentric casing strings and a radially innermost cement column of the plurality of cement columns, wherein the window defines a first axial length through the radially innermost casing string and a second axial length through the radially innermost cement column, and wherein the second axial length is greater than the first axial length; a primary plug set within an interior of the radially innermost casing string below the window; and a solidified secondary plug filling the window above the primary plug.

12. The wellbore repair system of claim 11, wherein the window is further defined radially through a casing string radially outward of the radially innermost casing string to a third axial length and through a cement column radially outward of the radially innermost cement column to a fourth axial length.

13. The wellbore repair system of claim 12, wherein the second axial length is greater than that of the third axial length and the fourth axial length, and wherein the first axial length is greater than that of the third axial length.

14. A wellbore system, comprising: a plurality of concentric casing strings disposed within a wellbore; a plurality of cement columns disposed radially outward of each of the concentric casing strings; a primary plug set within an interior of a radially innermost casing string of the plurality of concentric casing strings; a laser milled window defined in the radially innermost casing string and defining a first axial length; an abrasive jetting tool including a jetting head and one or more nozzles, couplable to the coiled tubing, and insertable within the flowpath of the wellbore to clean out cement from the radially innermost cement column to a second axial length greater than the first axial length by jetting an abrasive fluid behind the radially innermost casing string through the laser milled window; and a secondary plug material insertable into the window to generate a solidified secondary plug.

15. The wellbore system of claim 14, further comprising a window defined radially through at least the radially innermost casing string to a first axial length.

16. The wellbore system of claim 15, wherein the window is further defined through the radially innermost cement column to a second axial length, and wherein the second axial length is greater than the first axial length.

17. The wellbore system of claim 14, wherein the laser milling tool further includes: a reflector within the nozzle and oriented to redirect a laser beam from the coiled tubing towards an outlet end of the nozzle; and a focus lens interposing the outlet end of the nozzle and the reflector, and operable to focus the laser beam through the outlet end of the nozzle.

18. The wellbore system of claim 14, further comprising: a fluid reservoir stored at an external location; a pump in fluid communication with the fluid reservoir; and a fluid line in fluid communication with the fluid reservoir and the coiled tubing or the flowpath of the wellbore, wherein the fluid reservoir stores fluid for use in laser milling or abrasive jetting operations.
19. The wellbore system of claim 18, wherein the fluid is an optical fluid for flushing of the wellbore and transmission of a laser beam therethrough.
20. The wellbore system of claim 18, wherein the fluid is an abrasive fluid for cleaning out cement via the abrasive jetting tool.